

# **Dietary Carbohydrate Quality and Physical Functioning**

## **The New England Centenarian Study**

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Project Summary

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### **Project Overview**

This project examined the relationship between dietary carbohydrate quality and physical function in older adults using data from the New England Centenarian Study (NECS). The goal was to derive participant-level carbohydrate quality measures from food frequency questionnaire (FFQ) data and assess their associations with Instrumental Activities of Daily Living (IADL) scores using longitudinal statistical models.

Datasets harmonized for this analysis included the FFQ, nutrients data, cereal data, demographic data, and IADL outcome data. The final analytic sample was  $N = 438$  participants after applying exclusion criteria.

### **Data Processing and Exposure Development**

*Cereal Standardization*

The FFQ included cereal responses as free text rather than standardized categories. These were systematically reviewed and assigned to one of eleven cereal groups based on composition and processing characteristics:

- Refined flake cereal
- Sweetened flake cereal
- Whole oat cereal
- Sweetened oat cereal
- Bran cereal
- Wheat-based shredded cereal
- Granola/muesli
- High-fiber cereal
- Unspecified low-GI, unspecified medium-GI, and unspecified high-GI

Glycemic index (GI) and glycemic load (GL) values were assigned to each group using three main reference sources, anchored primarily in data from the University of Sydney Glycemic Index Research Database. A mean GI was calculated for each group from available reference data and applied uniformly to cereals assigned to that category. The resulting cereal dataset included participant-level average GI and GL values, and was merged with the full FFQ database prior to participant-level calculations.

#### *Exposure Variable Construction*

For each food item in the FFQ, daily carbohydrate intake was computed, GI values were assigned, and GL was calculated as  $GL = (GI \times \text{carbohydrates}) / 100$ . Participant-level exposures were then derived:

- Dietary glycemic index: carbohydrate-weighted average GI across all foods
- Dietary glycemic load: sum of food-level GL values
- Total carbohydrate intake (g/day)
- Total dietary fiber intake (g/day)
- Fiber density: grams of fiber per 1,000 kcal
- Carbohydrate-to-fiber ratio: lower values indicate higher carb quality

#### *Data Cleaning and Quality Control*

Participants with implausible energy intakes (< 600 or > 4,200 kcal/day) and those missing sex were excluded. Distributions of all dietary variables were reviewed prior to modeling.

Participants with elevated GL values were examined at the food level and it was noted that elevated GL was attributable to high intake of fruit juice, refined grains, and baked goods, and therefore, no observations were removed on these grounds. All dietary variables, except fiber density, were energy-adjusted using the residual method prior to modeling.

#### **Outcome and Covariates**

Physical Functioning was assessed using a 14-item IADL scale ranging from 0-14, with higher scores reflecting better functioning. Participants generally entered the study near the top of the scale at an average of 13.81, reflecting the high-functioning nature of this cohort. IADL was assessed at multiple visits across the follow-up.

A time variable was constructed as: Time = year of visit - year of enrollment, aligning all participants to a common baseline and allowing interpretation of model coefficients as change in

IADL per year. Observations where IADL was missing at a given visit were dropped, but participants were retained if they had at least one observed IADL score.

Covariates included age, sex, education, history of cardiovascular disease, stroke, diabetes, and cancer, and smoking and alcohol use.

## Statistical Analysis

Longitudinal associations between each carbohydrate exposure and IADL were modeling using generalized estimating equations (GEE), with a normal distribution, identity link, and exchangeable working correlation structure. The subject cluster variable was participant ID. Each exposure was modeled separately across three progressive adjusted models:

- Model 1: time + exposure + age + sex + education
- Model 2: Model 1 + CVD + stroke + diabetes + cancer
- Model 3: Model 2 + smoking + alcohol use

Stratified analyses were conducted by sex, diabetes status, and by chronic disease burden, using the Model 3 specification within each stratum.

## Descriptive Statistics

***Table 1. Baseline Characteristics of NECS Participants***

### ***Continuous Variables***

| Variable                  | N   | Mean $\pm$ SD    | Min | Max |
|---------------------------|-----|------------------|-----|-----|
| Age at enrollment (years) | 438 | 71.16 $\pm$ 7.23 | 39  | 90  |
| Education (years)         | 438 | 15.81            | 8   | 26  |

| Variable                          | N   | Mean $\pm$ SD       | Min   | Max     |
|-----------------------------------|-----|---------------------|-------|---------|
| IADL score                        | 438 | 13.81               | 5     | 14      |
| Fiber density (g/1000 kcal)       | 437 | 11.49 $\pm$ 3.68    | 3.57  | 35.84   |
| Carbohydrate:fiber                | 437 | 10.98 $\pm$ 3.17    | 3.70  | 24.20   |
| Glycemic Index (average)          | 437 | 57.42 $\pm$ 3.67    | 41    | 70.43   |
| Glycemic Load (sum)               | 437 | 134.09 $\pm$ 70.52  | 5.72  | 768.92  |
| Total carbohydrate intake (g/day) | 437 | 233.96 $\pm$ 109.37 | 28.93 | 1321.76 |
| Fiber intake (g/day)              | 437 | 22.64 $\pm$ 11.99   | 3.85  | 128.04  |

### *Categorical Variables*

| Variable                            | N (%)      |
|-------------------------------------|------------|
| <b>Sex</b>                          |            |
| Female                              | 239 (54.6) |
| Male                                | 199 (45.4) |
| <b>Diabetes</b>                     |            |
| No                                  | 405 (92.5) |
| Yes                                 | 32 (7.7)   |
| <b>Cardiovascular disease (CVD)</b> |            |
| No                                  | 415 (94.8) |
| Yes                                 | 22 (5.0)   |
| <b>Stroke</b>                       |            |
| No                                  | 429 (98.0) |
| Yes                                 | 8 (1.8)    |
| <b>Cancer</b>                       |            |
| No                                  | 357 (81.5) |
| Yes                                 | 80 (18.3)  |
| <b>Smoking history</b>              |            |
| No                                  | 194 (44.3) |

| Variable                   | N (%)      |
|----------------------------|------------|
| Yes                        | 243 (55.5) |
| <b>Alcohol use</b>         |            |
| No                         | 424 (96.8) |
| Yes                        | 13 (3.0)   |
| <b>Any chronic disease</b> |            |
| No                         | 309 (70.6) |
| Yes                        | 128 (29.2) |

Participants were generally high-functioning at baseline, with high educational attainment and relatively low prevalence of chronic disease. Mean baseline IADL scores were near the upper limit of the scale, suggesting a relatively healthy aging cohort with limited functional impairment at study entry.

***Table 2. Distribution of Energy-Adjusted Dietary Carbohydrate Quality Variables***

| Variable                                     | Mean    | Std Dev | Min    | Max     |
|----------------------------------------------|---------|---------|--------|---------|
| Total carbohydrates, energy-adjusted (g/day) | 227.11  | 46.94   | 92.61  | 385.84  |
| Dietary fiber, energy-adjusted (g/day)       | 21.89   | 6.97    | 5.72   | 54.99   |
| Glycemic load, energy-adjusted               | 129.30  | 38.21   | 33.43  | 281.19  |
| Total energy intake (kcal/day)               | 1915.15 | 613.46  | 602.36 | 4119.42 |

Energy adjustment reduced the variability and range of dietary exposure variables, particularly for total carbohydrate intake and glycemic load, suggesting that extreme values in unadjusted measures were partially driven by differences in total caloric intake rather than dietary composition alone.

**Table 3. Pearson Correlations Among Dietary Carbohydrate Quality Variables**

|                    | Carbohydrates (adj) | Fiber (adj)   | Glycemic Load   |
|--------------------|---------------------|---------------|-----------------|
| Carbohydrates, adj | 1.00                | 0.45 (<.0001) | 0.72 (< 0.0001) |
| Fiber, adj         | -                   | 1.00          | 0.29 (< 0.0001) |
| Glycemic load, adj | -                   | -             | 1.00            |

As expected, total carbohydrate intake and glycemic load were strongly positively correlated ( $r = 0.72$ ). Fiber intake showed a moderate positive correlation with total carbohydrate intake ( $r = 0.45$ ) and a weaker correlation with glycemic load ( $r = 0.29$ ), suggesting that fiber-containing foods varied substantially in glycemic properties across the cohort.

## Results

### *Association with Overall IADL Level*

These models assess whether each dietary exposure is associated with average IADL functioning across follow-up.

**Table 4. Model 1 - Age, Sex, Education Adjusted**

| Exposure            | $\beta$ | p-value |
|---------------------|---------|---------|
| Fiber density       | -0.009  | 0.90    |
| Carbohydrate:fiber  | -0.05   | 0.18    |
| Glycemic index      | -0.007  | 0.80    |
| Glycemic load       | -0.004  | 0.001   |
| Total carbohydrates | -0.002  | 0.0008  |

In the base model, higher glycemic load and total carbohydrate intake were significantly associated with lower IADL scores.

**Table 4.1. Model 2 - Model 1 + CVD, Stroke, Diabetes, Cancer**

| Exposure            | $\beta$ | p-value  |
|---------------------|---------|----------|
| Fiber density       | 0.78    | < 0.0001 |
| Carbohydrate:fiber  | -0.23   | < 0.0001 |
| Glycemic index      | -0.11   | 0.15     |
| Glycemic load       | 0.008   | 0.19     |
| Total carbohydrates | 0.005   | 0.24     |

After adjusting for chronic disease, glycemic load and total carbohydrate intake lose significance, consistent with confounding in Model 1. Fiber density emerges as a strong positive predictor ( $\beta = 0.78$ ,  $p < .0001$ ), and carbohydrate-to-fiber ratio shows a significant negative association ( $\beta = -0.23$ ,  $p < .0001$ ). This suggests carbohydrate quality, and particularly fiber, is associated with better functional status independent of disease burden.

**Table 4.2. Model 3 - Model 2 + Smoking, Alcohol (Fully Adjusted)**

| Exposure            | $\beta$ | p-value |
|---------------------|---------|---------|
| Fiber density       | 0.70    | <.0001  |
| Carbohydrate:fiber  | -0.19   | <.0001  |
| Glycemic index      | 0.08    | 0.50    |
| Glycemic load       | -0.03   | <.0001  |
| Total carbohydrates | -0.009  | 0.059   |

Fiber density ( $\beta = 0.70$ ,  $p < .0001$ ) and carbohydrate-to-fiber ratio ( $\beta = -0.19$ ,  $p < .0001$ ) remain significant in the fully adjusted model. Glycemic load re-emerges as significant here after being non-significant in Model 2. This is worth noting as a potential suppression effect from the lifestyle covariates. Overall, findings indicate that carbohydrate quality is associated with average functional status across follow-up.



## Stratified Analyses

**Table 5. Fully Adjusted Model Stratified by Sex**

| Exposure            | Female (N = 238) |      | Male (N = 199) |       |
|---------------------|------------------|------|----------------|-------|
|                     | $\beta$          | p    | $\beta$        | p     |
| Fiber density       | 0.04             | 0.45 | 0.16           | 0.11  |
| Carbohydrate:fiber  | -0.04            | 0.09 | -0.07          | 0.016 |
| Glycemic index      | 0.004            | 0.83 | -0.04          | 0.17  |
| Glycemic load       | -0.001           | 0.44 | -0.003         | 0.09  |
| Total carbohydrates | -0.0006          | 0.43 | -0.002         | 0.05  |

Sex-stratified analyses suggested potential differences in associations between carbohydrate quality measures and IADL function by sex. Among male participants, a higher carbohydrate-to-fiber ratio was significantly associated with lower IADL scores ( $\beta = -0.07$ ,  $p = 0.016$ ), with borderline associations observed for higher glycemic load ( $\beta = -0.003$ ,  $p = 0.09$ ) and total carbohydrate intake ( $\beta = -0.002$ ,  $p = 0.05$ ). In contrast, no significant associations were observed among female participants. Fiber density and glycemic index were not significantly associated with IADL function in either sex-stratified model.

**Table 6. Fully Adjusted Model Stratified by Diabetes Status**

| Exposure           | Diabetes (N = 32) |      | No Diabetes (N = 405) |       |
|--------------------|-------------------|------|-----------------------|-------|
|                    | $\beta$           | p    | $\beta$               | p     |
| Fiber density      | -0.29             | 0.40 | 0.08                  | 0.11  |
| Carbohydrate:fiber | -0.005            | 0.93 | -0.06                 | 0.012 |
| Glycemic index     | -0.05             | 0.31 | -0.004                | 0.80  |
| Glycemic load      | -0.005            | 0.40 | -0.001                | 0.14  |

| Exposure            | Diabetes (N = 32) |      | No Diabetes (N = 405) |      |
|---------------------|-------------------|------|-----------------------|------|
|                     | $\beta$           | p    | $\beta$               | p    |
| Total carbohydrates | -0.001            | 0.70 | -0.0009               | 0.09 |

No carbohydrate variable was significantly associated with IADL among participants with diabetes. Among those without diabetes, carbohydrate-to-fiber ratio showed a significant negative association ( $\beta = -0.06$ ,  $p = 0.012$ ). This could suggest carbohydrate quality matters more before metabolic disease has developed, though the diabetes stratum is small and likely underpowered.

**Table 7. Fully Adjusted Model Stratified by Any Chronic Disease**

| Exposure                  | Any Disease (N = 128) |      | No Disease (N = 309) |      |
|---------------------------|-----------------------|------|----------------------|------|
|                           | $\beta$               | p    | $\beta$              | p    |
| Fiber density             | 0.18                  | 0.21 | 0.05                 | 0.30 |
| Carbohydrate: fiber ratio | -0.12                 | 0.01 | -0.02                | 0.18 |
| Glycemic index            | -0.13                 | 0.50 | -0.006               | 0.80 |
| Glycemic load             | -0.002                | 0.30 | -0.001               | 0.19 |
| Total carbohydrates       | -0.0001               | 0.90 | -0.001               | 0.06 |

Among participants with any chronic disease, carbohydrate-to fiber ratio remained the only exposure significantly associated with IADL ( $\beta = -0.12$ ,  $p = 0.01$ ). No associations were significant in the disease-free group. This pattern suggests carbohydrate quality may have the most functional impact among those who are already metabolically or physiologically vulnerable. The null findings in the healthy group may also partly reflect ceiling effects in IADL score, as this is a high functioning cohort, especially among those without any chronic disease.

## Interpretation

Across models, carbohydrate quality measures, particularly those capturing dietary fiber, were more consistently associated with IADL functioning than total carbohydrate intake. The key takeaways include:

- Higher fiber density was associated with significantly better IADL functioning in both health-adjusted and fully adjusted models.
- Higher carbohydrate-to-fiber ratio (poorer carb quality) was consistently associated with lower IADL scores.
- Glycemic index showed no meaningful association across models.
- Glycemic load showed some inconsistency across model specifications, likely related to confounding and suppression by disease and lifestyle covariates.
- Total carbohydrate intake alone was not a strong predictor of functioning after adjustment.
- Stratified analyses suggest associations may be stronger among men than women.

Additionally, associations may be strongest among those with existing chronic disease, while diet-function relationships appear weaker in the disease-free subgroup, potentially due to the high-functioning nature of this cohort.

Overall, these findings support the interpretation that carbohydrate quality, not quantity, is the more functionally relevant dietary dimension in this population, and that dietary fiber is the primary driver of that association.

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